MIL-L-36304D
29 July 1977
SUPERSEDING
MIL-L-36304C
16 January 1975 and
MIL-L-36304B
29 August 1966

MILITARY SPECIFICATION

LENS MEASURING INSTRUMENT, OPHTHALMIC
110 Volt, 60 Hz, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of lens measuring instrument for use in analyzing and marking prescription lenses.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods of.
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-M-7298	-	Manual, Technical, Commercial Equipment.

Beneficial domments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code MTI, 2800 S. 20th St., Philadelphia, PA 19101, by using the self addressed Standardization Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6540

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained in the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INCORPORATED (ANSI)

Z80.1-1972 - American National Standard Requirements for First Quality Prescription Ophthalmic Lens.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

UNDERWRITERS' LABORATORIES, INC.

Underwriters' Laboratories Standards.

(Underwriters' Laboratories Standards are available from the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L. I., N. Y. 11749; 207 E. Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blvd., Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit to the contracting agency, prior to delivery, proof that the equipment he will supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the equipment conforms to this requirement.

3.1.2 In lieu of the label or listing, the contractor may submit, prior to delivery, independent proof satisfactory to the contracting agency that his equipment conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc.

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3.1.3 Compliance with Underwriters' Laboratories, Inc. requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Material. Material shall be of the highest quality throughout, and suitable in all respects for the purpose intended.

3.3 Performance characteristics. The lens measuring instrument shall be suitable for use on a 110 volt, 60 Hz, AC circuit, and shall be capable of determining the following optical parameters within tolerances specified in the American National Standards Institute (NSI) Z80.1-1972 for the ranges listed in table I. In additon, the instrument shall facilitate the optical center location, and measurement of prism power, prism power imbalance, addition power and marginal power (with modification) as required by above standard.

TABLE I. Optical parameter ranges.

Parameter	Range (minimum)
Sphere power	0 to ± 20 Diopters.
Cylinder power	0 to ± 20 Diopters.
Prism power	0 to 5 Diopters (prism).
Cylinder axis	0° to 180°
Prism axis	0° to 180°
Multifocal segment power	0 to + 20 Diopters.

3.4 Construction. Construction of the lens measuring instrument shall be sturdy and sufficiently wear-proof to withstand continuous usage for a five year period, seven hour day, in an optical laboratory, without impaired accuracy (see 4.5).

3.5 Component parts.

3.5.1 Spectacle holding table. The spectacle holding table shall be sturdily constructed, and designed for holding spectacles or a single lens in the instrument while being measured and analyzed. The table shall be used in conjunction with a spring loaded lens holder. The spectacle holding table shall be 6 to 6-3/4 inches in length and adjustable vertically through a range of 50 mm. minimum.

3.5.2 Lens holder. The lens holder shall be designed to hold the lens being analyzed (in or out of the spectacle frame) securely against the lens stop. The lens holder shall be fitted with a cushioning material (rubber or plastic) that will allow secure surface contact without scratching the lens.

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3.5.2 Lens holder. The lens holder shall be designed to hold the lens being analyzed (in or out of the spectacle frame) securely against the lens stop. The lens holder shall be fitted with a cushioning material (rubber or plastic) that will allow secure surface contact without scratching the lens.

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3.5.2.1 Lens stop. The lens stop shall be composed of a resilient plastic or other suitable material. The stop shall be securely mounted, and shall position the lens properly for accurate instrument measurements without scratching or otherwise damaging the lens.

3.5.3 Diopter power wheel. The diopter power wheel shall be conveniently located on the outside of the instrument and be of a suitable size for the operator's hand. The periphery shall be knurled. An accurate lens power scale shall be provided , conveniently located on the power wheel or superimposed on the instruments field of view within the eyepiece. This scale shall be in 1/8 diopter steps from 0 through ± 3.00 diopters, and in 1/4 diopter steps through ± 20 diopters.

3.5.4 Cylinder axis wheel. The cylinder axis wheel shall be conveniently located on the outside of the instrument. The wheel shall be calibrated in one degree increments from 0° to 180°. These graduations shall be sufficiently spaced for clarity and reading accuracy.

3.5.5 Target. The target shall consist of two sets of three parallel lines, discernible and centered in the field of vision of the instrument. One set shall be for spherical power measurements and the other for cylindrical power measurements. The cylinder portion shall consist of three wide parallel lines. The sphere portion shall consist of three narrow parallel lines at right angles to the wide lines. As an alternate, the sphere portion of the target shall consist of one wide and two narrow parallel lines at right angles to the wide lines of the cylinder portion. The wide line shall be the center line and shall be approximately the same width as one line of the cylinder portion. The target, target mount, target mount bearing and slide, and light source shall be totally enclosed in a suitable housing.

* 3.5.6 Eyepiece. An adjustable eyepiece shall be recessed or tapered with nylon ring or plastic foam eyepiece cushion to permit the use of the lens measuring instrument by individuals wearing glasses.

3.5.7 Prism measuring system. The prism measuring system shall consist of a holder, scale and a reticle.

3.5.7.1 Holder for auxiliary prisms. The holder for auxiliary prisms shall be located immediately in front of the telescope. The holder shall be designed for the smooth insertion and secure retention of auxiliary prisms.

3.5.7.2 Scale. The scale shall be calibrated in five degree increments, from 0° through 180°.

3.5.7.3 Reticle. The lens reticle shall be designed with a base-apex line and prism power rings, or other calibrations, indicating up to five prism diopters.

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3.5.8 Lens marking device. The lens marking device shall consist of three marking pens and an ink reservoir. The pens shall be capable of marking the optical center and the 180° axis of the lens in the same operation.

* 3.5.9 Light source. The light source shall provide sufficient illumination for proper operation of the instrument. The light source shall be completely enclosed and located to permit replacement of the lamp without difficulty. A readily removable cover shall provide access to the lamp. The instrument shall have a pilot indicator, which shall show when the lamp is illuminated. The lamp illumination shall be controlled by a switch, located conveniently on the instrument, with three positions, off, low for normal illumination and high for measuring absorptive lenses. As an alternate, the instrument shall have a swing out filter lens to control illumination and which swings into position for normal use and swings out of the optical path for measurement of darker lenses.

* 3.5.10 Housing and base. The instrument shall have a sturdy metal housing and a cast metal base. The base shall be constructed to insure stability, and to permit an adjustable vertical inclination of 0° to 60° for the instrument. The housing and base shall have a baked-on enamel finish.

3.5.11 Electric cord and plug. The electric cord and plug, provided with the instrument, shall be suitable for supplying current from a nominal 115 volt, single phase AC power source, and shall comply with the requirements of 3.1.

3.5.12 Accessories. The lens measuring instrument shall be furnished with two spare lamps, and one spare ink pad.

* 3.6 Identification marking. Each lens measuring instrument shall be permanently and legibly marked with the name or registered trademark of the manufacturer and serial number, when available.

3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298 MANUAL, TECHNICAL Commercial Equipment. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order.

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In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of the 15 additional copies shall be at the option of the Government.

3.8 Workmanship. Workmanship shall be first class throughout. Lens measuring instruments shall be free from defects which detract from their appearance or impair their serviceability. Instrument lenses shall be free of pits, scratches, bubbles, striae, inclusions and waves.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 5 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

* 4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of any component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be in accordance with inspection level II of MIL-STD-105, with an AQL of 4.0 percent defective.

4.2.2 Sampling for tests. Samples selected in accordance with 4.2.1 shall be tested as specified in 4.4. AQL shall be 2.5 percent defective.

4.3 Examination. The lens measuring instrument shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to, the classification of defects listed in table II. Additional defects shall be classified in accordance with MIL-STD-105.

TABLE II. Classification of defects.

Categories	
Major	
101	Any component missing or not as specified.
102	Any dimension not within specified tolerance.
103	Holding table not adjustable through specified range.
104	Calibration of diopter power wheel or cylindrical axis wheel incorrect.
105	Target lines indiscernible or not clearly designated for sphere and cylinder markings.
106	Component does not function as required.
107	Lenses not free of pits, scratches, bubbles, striae, inclusions and waves.
Minor	
201	Poor workmanship; e.g., sharp edges, burrs, poor finish.
202	Scales on prism increments other than specified.
203	Accessories, spare lamps or spare ink pad missing.
204	Marking illegible, incorrect or missing.

4.3.2 Packaging inspection. The inspection of the preservation packaging shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provision of the applicable container specification and the marking requirements of MIL-STD-129.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Performance characteristics. Capability of the instrument shall be determined in accordance with the procedures contained in the American National Standard Requirements for First-Quality Prescription Ophthalmic Lenses, American National Standards Institute Standard (ANSI) Z80.1-1972.

4.5 Certification. The manufacturer shall certify the instrument as sufficiently wear-proof for the five year period specified in 3.4. The certificate shall be conditioned on use of the instrument by trained operators, location in suitable environment and routine maintenance.

4.6 Inspection. Inspection shall be conducted by qualified personnel thoroughly familiar with optical equipment.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each instrument and accessories shall be packaged in die-cut polyurethane and further packaged in accordance with MIL-P-116, method 1C-2.

5.1.2 Level C. Instruments shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each instrument shall be packed in an exterior container designed for a type 2 level constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

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5.2.2 Level B. Each instrument shall be packed in an exterior container, designed for a type 2 load constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged instruments shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The instrument covered by this specification is used for determination of lens prescription, layout, marking, and verification of finished lens product.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).
(c) Applicable NSN.

6.3 NSN coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN 6540-00-181-8037	LENS MEASURING INSTRUMENT, OPHTHALMIC, 115 Volt, 60 Hz, AC
NSN 6540-00-299-8134	LENS MEASURING INSTRUMENT, OPHTHALMIC, 115 Volt, 60 Hz, AC .

6.4 Changes from previous issues. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD-MB

Preparing activity:
DLA-DM

Project Number 6540-1046

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE	
MIL-L-36304D	LENS MEASURING INSTRUMENT, OPHTHALMIC, 110 Volt, 60 Hz, AC

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
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